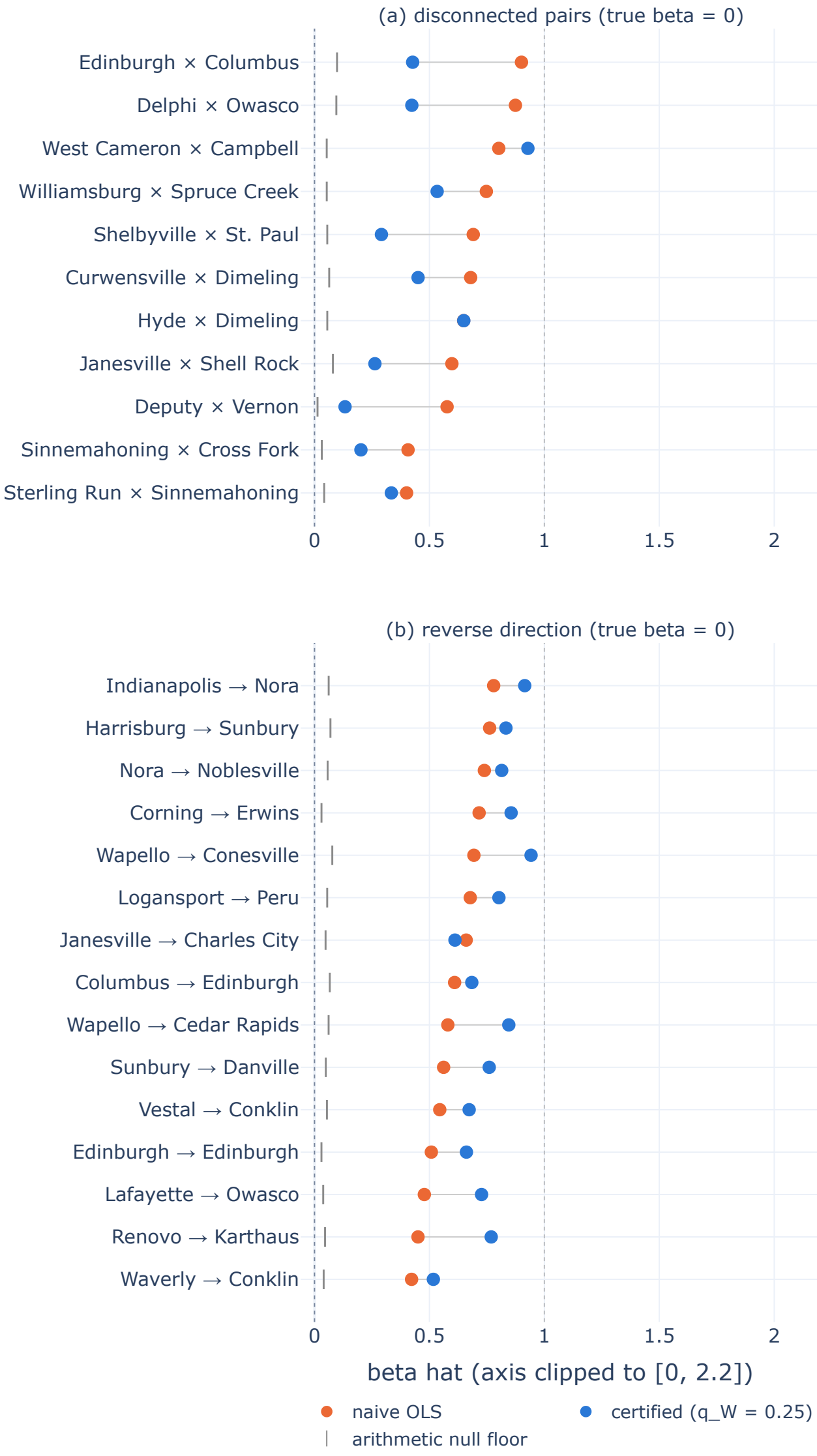
(a) disconnected pairs (true beta = 0)
Edinburgh × Columbus
Delphi × Owasco
West Cameron × Campbell
Williamsburg × Spruce Creek
Shelbyville × St. Paul
Curwensville × Dimeling
Hyde × Dimeling
Janesville × Shell Rock
Deputy × Vernon
Sinnemahoning × Cross Fork
Sterling Run × Sinnemahoning
0
0.5
1
1.5
2
(b) reverse direction (true beta = 0)
Indianapolis → Nora
Harrisburg → Sunbury
Nora → Noblesville
Corning → Erwins
Wapello → Conesville
Logansport → Peru
Janesville → Charles City
Columbus → Edinburgh
Wapello → Cedar Rapids
Sunbury → Danville
Vestal → Conklin
Edinburgh → Edinburgh
Lafayette → Owasco
Renovo → Karthaus
Waverly → Conklin
0
0.5
1
1.5
2
beta hat (axis clipped to [0, 2.2])
naive OLS
certified (q_W = 0.25)
arithmetic null floor